

Google Meeting Agent System

Statement of Work: AI Meeting Assistant POC

Table of Contents

1. [Introduction](#)
 - 1.1 [Background](#)
 - 1.2 [Objectives and Proposed Solution](#)
 - 1.3 [System Overview](#)
 2. [Project Success Criteria](#)
 3. [Scope of Work](#)
 4. [Assumptions](#)
 5. [Functional Requirements](#)
-

1. Introduction

1.1 Background

Organizations face a critical challenge when non-technical team members (such as sales representatives, account managers, or project coordinators) conduct client meetings without the presence of solution architects or technical experts. Key pain points include:

Knowledge Gap During Live Meetings:

- Non-technical staff are unable to answer technical questions from clients in real-time
- Absence of solution architects or domain experts creates awkward pauses and delays
- Inability to provide immediate, accurate responses damages client confidence and credibility
- Follow-up emails and delayed responses slow down sales cycles and decision-making processes
- Critical technical questions may be misunderstood or inadequately addressed

Lack of Continuous Improvement:

- No systematic way to review meeting performance and identify missed opportunities
- Teams cannot analyze what questions were asked, how they were answered, and what could have been improved
- Knowledge gaps repeat across multiple meetings because there's no feedback loop
- New team members lack access to historical context and best-practice responses
- Overall team quality stagnates without data-driven insights into meeting effectiveness

Resource Allocation Challenges:

- Solution architects and technical experts cannot attend every client meeting
- Organizations struggle to scale technical expertise across multiple concurrent meetings
- High-value technical resources spend time on routine questions that could be automated
- Geographic differences make expert availability even more challenging

These challenges result in lost sales opportunities, reduced client satisfaction, slower team development, and inefficient use of technical expertise. Organizations need a solution that bridges the knowledge gap during live meetings while providing retrospective analysis to drive continuous improvement in team performance and response quality.

1.2 Objectives and Proposed Solution

The objective of this solution is to empower non-technical team members with AI-powered technical expertise during client meetings, while providing systematic retrospective analysis to drive continuous team improvement.

Primary Objectives:

1. Bridge the Knowledge Gap in Real-Time:

- Enable non-technical staff to have immediate access to technical expertise during client meetings
- Provide accurate, contextual answers to technical questions without requiring solution architects to be present
- Maintain client confidence and meeting momentum by having the ability to provide swift answers.

2. Enable Continuous Team Improvement:

- Capture all questions and responses for post-meeting analysis
- Allow teams to review what was asked, how it was answered, highlight if key agenda items were not discussed, and identify improvement opportunities,
- Provide coaching insights for team members to improve their meeting performance (What was good?, what could be better?, Score rating on rubrics)

3. Scale Technical Expertise:

- Allow solution architects and technical experts to contribute knowledge without attending every meeting
- Support multiple concurrent meetings with consistent technical response quality
- Reduce dependency on high-value technical resources for routine questions

Solution Capabilities:

The solution will provide:

- **AI Meeting Agents** that join Google Meet sessions and monitor conversations in real-time
- **Live Assistance Mode** where agents provide real-time answers to detected questions by the client in question answer pairs, enabling support during active meetings
- **Retrospective Analysis Mode** where teams review meeting transcripts, identify improvement opportunities, highlight any missed agenda items, and summarise the action items/next steps.
- **Customizable Agent Personalities** allowing different agents to match different meeting contexts
- **Admin Configuration Portal** for solution architects to manage knowledge bases and tune agent behavior, create, update or delete agents.
- **User Monitoring Portal** for meeting participants to access live assistance and conduct post-meeting reviews

This solution transforms every meeting into a learning opportunity while ensuring non-technical team members can confidently handle technical discussions with clients.

The solution will be designed as a cloud-based application with separate admin and user portals, leveraging modern speech-to-text streaming services and AI language models for intelligent meeting participation.

1.3 System Overview

The system consists of the following major components:

Meeting Agent Engine:

- Automated Google Meet session joining using Gmail authentication
- Real-time audio capture and speech-to-text processing
- Question detection and natural language understanding
- Response generation using configured knowledge bases and system instructions

Admin Portal:

- Agent configuration and management interface
- System instruction and prompt tuning capabilities
- Knowledge base creation and document upload
- Gmail account management for multiple agents (Assign a gmail account credentials for agent to auth)
- Question-answer pair monitoring and analytics
- Role-based access control for administrators

User Portal:

- Real-time meeting transcription viewing (Live Output Mode)
- Historical question-answer pair review (Retro Mode)
- Agent status and connection monitoring
- Search and filtering capabilities for past interactions
- Role-based access control for standard users

Data Flow:

1. Meeting Agent joins Google Meet session using Gmail credentials
 2. Audio stream is captured and processed through speech-to-text service
 3. Transcribed text is analyzed for questions using natural language processing
 4. When a question is detected, the agent queries the configured knowledge base
 5. Response is generated using system instructions and knowledge base context
 6. Response is delivered as audio in the meeting and logged to admin portal
 7. All question-answer pairs are stored for retrospective analysis
 8. Users can view real-time activity or review historical interactions through the user portal
-

2. Project Success Criteria

The solution will be considered successful when the following criteria are met:

Agent Participation:

- Meeting agents shall successfully authenticate and join Google Meet sessions using Gmail credentials with 95% success rate
- Agents shall maintain stable connections throughout meeting duration with less than 1% disconnection rate

Question Detection and Response:

- The system shall detect questions in meeting audio with minimum 80% accuracy based on test scenarios
- The system shall generate contextually relevant responses with minimum 80% relevance rating from evaluators
- Response latency shall not exceed 10 seconds from question detection to audio delivery

Real-Time Performance:

- Question identification shall display in user portal within 3 seconds of speech
- Agent responses shall appear in user portal within 1 second of generation
- System shall maintain 99% uptime during active meeting sessions

Administrative Capabilities:

- Administrators shall be able to create and customize system prompts to define unique agent personalities and objectives
- Administrators shall be able to configure knowledge bases for domain-specific responses
- Administrators shall be able to add, modify, and remove agent instances
- All question-answer pairs shall be logged and accessible in admin portal with 100% capture rate
- Token usage and audit trails should go to the Axrail admin portal for monitoring.

User Experience:

- Users shall be able to toggle between live and retro mode
- Users shall be able to search and filter historical question-answer pairs
- System shall support concurrent access by multiple administrators and users

Security and Access Control:

- Gmail credentials shall be encrypted at rest using industry-standard encryption
 - Role-based access control shall prevent unauthorized access to admin features
 - Dual portal interfaces shall enforce proper separation between admin and user capabilities
 - System shall implement account lockout after 3 failed authentication attempts
-

3. Scope of Work

The scope of work for this solution includes:

Infrastructure and Deployment:

- Provision cloud-based infrastructure to support the meeting agent system
- Deploy admin portal with agent configuration and management capabilities
- Deploy user portal with live and retrospective viewing modes
- Implement secure credential storage and encryption mechanisms

Meeting Agent Development:

- Develop automated Google Meet joining capability using Gmail authentication
- Integrate speech-to-text service for real-time audio transcription
- Implement question detection using natural language processing
- Develop response generation using configurable knowledge bases and system prompts
- Implement audio output delivery to meeting sessions

Admin Portal Features:

- Agent management interface (add, edit, delete agents)
- System prompt configuration with support for unique agent personalities and objectives
- Knowledge base management with document upload support
- Question-answer pair monitoring dashboard
- Administrator authentication and authorization
- Dual portal interface with role-based access control

User Portal Features:

- Live output mode with real-time transcription streaming
- Retro mode with historical question-answer pair display
- Mode toggle functionality with session persistence
- Search and filtering capabilities
- User authentication and authorization
- Dual portal interface with appropriate user permissions

Testing and Validation:

- Conduct testing with up to 50 representative meeting scenarios
- Validate question detection accuracy and response relevance
- Performance testing for real-time streaming and latency

- Security testing for credential management and access control
- User acceptance testing with stakeholders

Documentation and Knowledge Transfer:

- System architecture documentation
 - Administrator user guide
 - End-user guide
 - API documentation (if applicable)
 - Handover session with technical team
-

4. Assumptions

Infrastructure and Environment:

- The solution will be built entirely on Amazon Web Services (AWS) cloud platform, leveraging its scalable, secure, and reliable infrastructure. The high-level system architecture consists of the following components:

Google Meet Integration:

- Gmail accounts for meeting agents will be provided by the client or created specifically for this solution

Data and Content:

- All content and interactions will be in English only
- Client will provide up to 50 representative test scenarios for validation

Scope Limitations:

- This solution does not include integration with external CRM, ERP, or ticketing systems
 - This solution does not include mobile application development
 - This solution does not include support for meeting platforms other than Google Meet
 - This solution does not include multi-language support (English only)
-

5. Functional Requirements

5.1 Agent Meeting Participation

User Story: As an administrator, I want agents to join Google Meet sessions using Gmail accounts, so that they can participate in meetings automatically.

Acceptance Criteria

1. WHEN an administrator provides valid Gmail credentials, THE Meeting_Agent SHALL authenticate with Google Meet
2. WHEN a meeting link is provided, THE Meeting_Agent SHALL join the specified Google Meet session
3. WHILE connected to a meeting, THE Meeting_Agent SHALL maintain an active connection
4. IF authentication fails, THEN THE Meeting_Agent SHALL log the error and notify the administrator
5. WHEN the meeting ends, THE Meeting_Agent SHALL disconnect gracefully

5.2 Speech Recognition and Question Detection

User Story: As a user, I want the agent to monitor meeting audio for questions, so that it can provide timely responses.

Acceptance Criteria

1. WHERE Live_Output_Mode is enabled, THE Meeting_Agent SHALL process audio through the Speech_To_Text_Stream
2. WHEN audio is transcribed, THE Meeting_Agent SHALL analyze the text for questions
3. WHEN a question is detected, THE Meeting_Agent SHALL generate a response based on configured Knowledge_Base and System_Instructions if available.
4. WHEN no question is detected, THE Meeting_Agent SHALL continue monitoring without responding
5. THE Meeting_Agent SHALL process transcription with a latency of less than 3 seconds from speech to detection

5.3 Response Delivery and Logging

User Story: As an administrator, I want all question-answer pairs logged to the admin portal, so that I can review agent interactions.

Acceptance Criteria

1. WHEN the Meeting_Agent generates a response, THE Meeting_Agent SHALL deliver the response during the Google Meet session in the Admin_Portal.
2. WHEN a Question_Answer_Pair is created, THE Meeting_Agent SHALL send it to the Admin_Portal
3. THE Admin_Portal SHALL store all Question_Answer_Pairs with timestamps and meeting context
4. THE Admin_Portal SHALL display Question_Answer_Pairs in chronological order
5. WHEN a response fails to deliver, THE Meeting_Agent SHALL log the failure and retry once

5.4 Admin Portal - Agent Configuration

User Story: As an administrator, I want to configure system prompts for each agent to define unique personalities and objectives, so that different agents can serve different purposes and contexts.

Acceptance Criteria

1. THE Admin_Portal SHALL allow administrators to create custom system prompts (e.g. agent objectives and behavioral guidelines, personality traits and answering tone) for each Meeting_Agent and can reference Knowledge_Base in the prompt.
2. WHEN system prompts are updated, THE Admin_Portal SHALL apply changes to the Meeting_Agent within 5 seconds
3. THE Admin_Portal SHALL validate system prompts before saving
4. THE Admin_Portal SHALL maintain a version history of system prompt changes with timestamps
5. THE Admin_Portal SHALL allow administrators to preview how different prompts affect agent responses

5.5 Admin Portal - Knowledge Base Management

User Story: As an administrator, I want to manage knowledge bases for agents, so that they can answer domain-specific questions accurately.

Acceptance Criteria

1. THE Admin_Portal SHALL allow administrators to create Knowledge_Base entries
2. THE Admin_Portal SHALL allow administrators to upload documents to a Knowledge_Base
3. THE Admin_Portal SHALL allow administrators to assign Knowledge_Base collections to specific agents
4. WHEN a Knowledge_Base is updated, THE Admin_Portal SHALL reindex the content within 30 seconds

5. THE Admin_Portal SHALL support document formats including PDF, TXT, and DOCX

5.6 Admin Portal - Agent Management

User Story: As an administrator, I want to add and manage multiple agents, so that I can deploy agents across different meetings and contexts.

Acceptance Criteria

1. THE Admin_Portal SHALL allow administrators to register new Meeting_Agent instances with Gmail credentials
2. THE Admin_Portal SHALL display a list of all configured Meeting_Agent instances
3. THE Admin_Portal SHALL allow administrators to enable or disable individual Meeting_Agent instances
4. THE Admin_Portal SHALL allow administrators to delete Meeting_Agent instances
5. WHEN a Meeting_Agent is deleted, THE Admin_Portal SHALL remove all associated credentials securely

5.7 User Portal - Output Mode Selection

User Story: As a user, I want to toggle between live and retro modes, so that I can choose how I monitor agent activity.

Acceptance Criteria

1. THE User_Portal SHALL provide a toggle control for switching between Live_Output_Mode and Retro_Mode
2. WHEN Live_Output_Mode is selected, THE User_Portal SHALL stream real-time transcription and agent responses
3. WHEN Retro_Mode is selected, THE User_Portal SHALL display historical Question_Answer_Pairs
4. THE User_Portal SHALL persist the user's mode selection across sessions
5. WHEN switching modes, THE User_Portal SHALL update the display within 1 second

5.8 User Portal - Live Output Display

User Story: As a user, I want to see real-time agent responses to questions, so that I can reference them for quick answers.

Acceptance Criteria

1. WHILE in Live_Output_Mode, THE User_Portal SHALL use Speech_To_Text_Stream output to transcribe meeting in real-time. Store the transcription for use by Live_Output or Retro_Mode
2. WHEN a question is detected, THE User_Portal SHALL generate and display the Question_Answer_Pairs

3. THE User_Portal SHALL auto-scroll to show the most recent content
4. THE User_Portal SHALL display connection status for the Meeting_Agent

5.9 User Portal - Retro Mode Display

User Story: As a user, I want to conduct a meeting retrospective, so that I can identify missed agenda items or improvements for my sales pitch.

Acceptance Criteria

1. WHILE in Retro_Mode, THE User_Portal SHALL process transcription and create a summary after completed meeting.
2. THE User_Portal SHALL display the full transcription for reference
3. The User_Portal SHALL generate areas for improvement as well as missed agenda topics or conversations by the user.
4. THE User_Portal SHALL allow users to chat and get insights based off the transcription about the meeting and performance.
5. THE User_Portal SHALL display meeting metadata including duration and participant count

5.10 Authentication and Authorization

User Story: As a system administrator, I want role-based access control with dual portal interfaces, so that administrators and users have appropriate permissions and access to their respective features.

Acceptance Criteria

1. THE System SHALL provide separate Admin_Portal and User_Portal interfaces
2. THE Admin_Portal SHALL require administrator authentication before granting access
3. THE User_Portal SHALL require user authentication before granting access
4. THE System SHALL prevent users from accessing Admin_Portal features
5. THE System SHALL allow administrators to access both Admin_Portal and User_Portal
6. THE System SHALL enforce role-based permissions at the API level
7. THE System SHALL log all authentication attempts and access violations

5.11 Gmail Credential Security

User Story: As an administrator, I want Gmail credentials stored securely, so that agent accounts are protected from unauthorized access.

Acceptance Criteria

1. WHEN Gmail credentials are provided, THE Admin_Portal SHALL encrypt them before storage
2. THE System SHALL use industry-standard encryption algorithms for credential storage

3. THE System SHALL never display Gmail passwords in plain text
4. THE System SHALL use secure credential storage mechanisms provided by the hosting environment
5. WHEN credentials are retrieved for authentication, THE System SHALL decrypt them in memory only

5.12 Real-Time Streaming Performance

User Story: As a user, I want minimal latency in live output, so that I can follow meeting conversations effectively.

Acceptance Criteria

1. WHILE in Live_Output_Mode, THE User_Portal SHALL display transcription within 2 seconds of speech
2. WHEN the Meeting_Agent generates a response, THE User_Portal SHALL display it within 1 second of generation
3. THE Speech_To_Text_Stream SHALL maintain a connection uptime of at least 99% during active meetings
4. IF the Speech_To_Text_Stream connection drops, THEN THE Meeting_Agent SHALL attempt reconnection within 5 seconds
5. THE User_Portal SHALL display a latency indicator showing current stream delay